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LITERATURE SURVEY

LEARNING GROUP - 5

TITLE OF THE PROJECT: REAL-TIME FAKE JOB OFFER DETECTION USING NATURAL LANGUAGE PROCESSING AND BROWSER EXTENSION INTEGRATION

PROJECT DOMAIN: NATURAL LANGUAGE PROCESSING

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TITLE OF THE PROJECT

REAL-TIME FAKE JOB OFFER DETECTION USING NATURAL LANGUAGE PROCESSING AND BROWSER EXTENSION INTEGRATION

1. ABSTRACT:

The increasing dependence on digital platforms for job searching has led to a significant rise in recruitment-related fraud. Fake job offers distributed through email services, professional networking platforms, and online job portals often deceive job seekers by demanding registration fees, processing charges, or sensitive personal information. Many users, particularly fresh graduates and unemployed youth, struggle to distinguish between legitimate opportunities and fraudulent recruitment messages due to the realistic language used by scammers.

To address this issue, this project proposes a real-time fake job detection system based on Natural Language Processing integrated within a browser extension environment. The system automatically analyzes job-related textual content from platforms such as Gmail, LinkedIn, and jobs portals, and predicts the likelihood of fraudulent intent using machine learning-based text classification techniques.

Text preprocessing techniques including tokenization, text normalization, stop-word removal, and vectorization are applied to prepare the data for modeling. A baseline classifier using traditional machine learning

algorithms such as Logistic Regression or SVM is developed for comparison, followed by a transformer-based model for improved contextual understanding.

The proposed solution provides a practical, deployable, and real-time fraud detection framework that enhances online safety for job seekers by identifying deceptive recruitment messages before users take financial or personal risks.

Keywords: Fake Job Detection, Recruitment Scam, Natural Language Processing, Text Classification, Machine Learning, BERT, Browser Extension, Real-Time Detection, Fraud Prevention

2. LITERATURE REVIEW OF PAPERS:

AUTHORS OF THE PAPER	OBJECTIVE	METHODOLOGY	PERFORMANCE / KEY OUTCOMES
Aashir Amaar et al. (2022) [1]	To detect fraudulent job advertisements using machine learning and NLP techniques while addressing class	Utilized the Kaggle fake job postings dataset. Applied text preprocessing including tokenization, stemming,	Extra Trees Classifier with TF-IDF and ADASYN achieved the highest accuracy (~99.9%). ADASYN significantly

	imbalance in the dataset.	punctuation removal, and normalization. Extracted features using Bag-of-Words (BoW) and TF-IDF. Addressed dataset imbalance using ADASYN oversampling. Implemented six supervised classifiers: Logistic Regression, Random Forest, KNN, SVM, Extra Trees, and MLP. Performance evaluated using Accuracy, Precision, Recall, and F1-score.	improved minority class detection. The study demonstrated that feature extraction and imbalance handling critically impact model performance.

Sokratis Vidros et al. (2017) *Automatic Detection of Online Recruitment Frauds*	To analyze Online Recruitment Fraud (ORF) and provide the first publicly available dataset for employment scam detection.	Introduced the Employment Scam Aegean Dataset (EMSCAD) consisting of 17,880 annotated job advertisements. Applied Bag-of-Words feature representation. Used machine learning classifiers including Logistic Regression, Random Forest, Naïve Bayes, and Decision Trees. Conducted exploratory data analysis and empirical evaluation.	Logistic Regression and Random Forest showed strong performance in detecting fraudulent job ads. The study highlighted dataset imbalance challenges and emphasized that text-only classification provides limited but foundational detection capability. Introduced a benchmark dataset widely

			used in later research.
Dinh-Hong Vu et al. (2025) *Improving Fake Job Description Detection Using Deep Learning-Based NLP Techniques*	To improve fake job description detection using deep learning-based NLP techniques by leveraging word embeddings and metadata integration.	Proposed NLP2FJD framework. Utilized pre-trained Word2Vec embeddings to extract semantic features from textual data. Combined textual features with meta-information from the EMSCAD dataset. Implemented deep learning models for classification. Conducted ROC curve analysis to determine optimal	NLP2FJD outperformed traditional machine learning methods. Integration of word embeddings and metadata significantly improved predictive performance. Achieved higher AUC and improved discrimination between fake and real job descriptions.

		classification threshold.	
Shashikant Kumar et al. (2025) *The Growing Threat of Fake Job Postings: A Review of Machine Learning and NLP-Based Detection Approaches*	To review and compare machine learning, deep learning, and NLP-based approaches for fake job post detection and identify research gaps.	Conducted a comparative literature survey of existing fake job detection models. Analyzed approaches including Bag-of-Words, TF-IDF, word embeddings, transformer models, ensemble classifiers, and oversampling techniques. Reviewed EMSCAD dataset usage and performance benchmarks.	Identified that transformer-based and embedding-based models generally outperform traditional ML models. Highlighted challenges such as class imbalance, dataset limitations, and need for real-time fraud detection systems. Emphasized requirement for

			larger and cleaner datasets.
Priya Khandagale et al. (2022) *Fake Job Detection Using Machine Learning*	To develop an automated machine learning-based classification system to distinguish between genuine and fraudulent job postings.	Applied supervised machine learning classification techniques. Conducted exploratory data analysis on job posting dataset. Implemented multiple classifiers and ensemble approaches to compare performance. Selected best-performing model based on accuracy evaluation.	Demonstrated that ensemble-based classifiers improved prediction accuracy compared to single classifiers. Confirmed that supervised learning models can effectively classify fake and legitimate job advertisements.

3. FINDINGS /INSIGHTS FROM THE LITERATURE SURVEY:

Observations from Paper 1:

This paper provides a strong machine learning-based baseline for fake job detection. The major contribution lies in addressing the severe class imbalance problem in the dataset using ADASYN. It demonstrates that traditional ML models can achieve very high accuracy when proper feature extraction (TF-IDF) and oversampling techniques are applied. The key takeaway is that dataset balancing and feature engineering significantly influence classification performance.

Existing Methods and Performance in Paper 1:

- Algorithms used: Logistic Regression, Random Forest, KNN, SVM, Extra Trees, MLP
- Feature extraction: Bag-of-Words (BoW), TF-IDF
- Dataset: Kaggle Fake Job Postings Dataset (~17,000 records)
- Oversampling: ADASYN
- Evaluation metrics: Accuracy, Precision, Recall, F1-score
- Extra Trees + TF-IDF + ADASYN achieved ~99.9% accuracy
- ADASYN improved minority class detection

Limitations:

- Heavy dependence on oversampling techniques
- Extremely high accuracy may indicate overfitting risk
- Traditional ML models lack deep semantic understanding

- No real-time deployment architecture

Improvisation of Methodology in Paper 1:

Future work could integrate contextual language models (BERT/transformers) instead of TF-IDF to capture semantic meaning. Real-time browser-based implementation and reduction of oversampling dependency using cost-sensitive learning could improve robustness.

Observations from Paper 2:

This paper is foundational as it introduced the EMSCAD dataset, which became the benchmark dataset for fake job detection research. It highlights the complexity of Online Recruitment Fraud (ORF) and explains that employment scam detection shares similarities with spam and phishing detection. The key insight is that text-only models provide limited detection power and that fraud detection requires contextual understanding.

Existing Methods and Performance in Paper 2:

- Dataset introduced: EMSCAD (17,880 job ads; highly imbalanced)
- Feature extraction: Bag-of-Words
- Algorithms used: Logistic Regression, Random Forest, Naïve Bayes, Decision Trees
- Evaluation: Empirical ML comparison

- Logistic Regression and Random Forest showed strong detection capability
- Highlighted dataset imbalance issue

Limitations:

- Heavy class imbalance (few fraudulent samples)
- Relied mainly on text-based features
- Limited deep learning exploration
- No advanced semantic modeling

Improvisation of Methodology in Paper 2:

Future research could incorporate word embeddings, contextual transformers, metadata fusion, and real-time detection mechanisms instead of relying solely on Bag-of-Words representation.

Observations from Paper 3:

This paper advances the field by integrating deep learning and semantic word embeddings (Word2Vec). It shows that combining textual information with metadata significantly enhances detection performance. The study emphasizes that contextual representation improves predictive ability compared to traditional ML methods.

Existing Methods and Performance in Paper 3:

- Framework: NLP2FJD
- Feature extraction: Pre-trained Word2Vec embeddings
- Dataset: EMSCAD
- Combined textual + meta-information
- Deep learning models used for classificationEvaluation: ROC analysis, AUC
- Achieved higher AUC compared to traditional ML approaches

Limitations:

- Used Word2Vec (static embeddings) instead of contextual embeddings like BERT
- Computationally heavier than traditional ML
- Still dependent on EMSCAD dataset

Improvisation of Methodology in Paper 3:

Future improvements could include transformer-based models (BERT, RoBERTa), multilingual detection, real-time inference systems, and browser-based implementation for practical deployment.

Observations from Paper 4:

This is a review paper that compares ML, deep learning, and NLP approaches. It highlights that transformer and embedding-based

approaches outperform traditional ML methods. The paper strongly emphasizes the need for real-time detection systems and larger, cleaner datasets. The key insight is that current research lacks scalable and deployable solutions.

Existing Methods and Performance in Paper 4:

- Compared ML, Deep Learning, NLP approaches
- Reviewed Bag-of-Words, TF-IDF, Word Embeddings, Transformer models
- Identified oversampling techniques (SMOTE, ADASYN) usage
- Observed transformer-based models achieving higher F1-scores
- Highlighted imbalance as recurring issue

Limitations:

- Review-based; no new experimental framework
- Focused mainly on EMSCAD dataset
- Lack of deployment perspective

Improvisation of Methodology in Paper 4:

Future work should focus on real-time detection systems (browser extensions, APIs), cross-platform integration, multilingual fraud detection, and evolving scam pattern detection.

Observations from Paper 5:

This paper presents a practical ML-based classification system comparing multiple supervised models. It emphasizes ensemble approaches for improved prediction accuracy. The key takeaway is that supervised learning can effectively distinguish fake and legitimate job ads when sufficient labeled data is available.

Existing Methods and Performance in Paper 5:

- Approach: Supervised ML classification
- Used multiple classification algorithms
- Compared single vs ensemble classifiers
- Evaluation: Accuracy-based comparison
- Ensemble models performed better than single classifiers

Limitations:

- Limited focus on advanced NLP techniques
- No deep learning models explored
- No imbalance handling discussion
- No real-time deployment

Improvisation of Methodology in Paper 5:

Future work could integrate contextual embeddings, imbalance-aware training, real-time browser extension deployment, and explainable AI components.

4. DISCUSSION:

- Across all the reviewed papers, there is strong agreement that machine learning and NLP-based classification is an effective approach for detecting fake job postings, significantly outperforming manual verification and rule-based filtering systems.
- Traditional machine learning classifiers such as Logistic Regression, Random Forest, SVM, and Extra Trees emerge as the most commonly used baseline models, particularly in Papers by Aashir Amaar et al., Sokratis Vidros et al., and Priya Khandagale et al., due to their simplicity, interpretability, and strong performance on structured text features like TF-IDF and Bag-of-Words.
- While traditional ML models are widely adopted, multiple studies highlight their limitations, particularly their inability to capture deep semantic context in textual data, which motivates the use of advanced techniques such as Word Embeddings and Deep Learning models (as demonstrated in Dinh-Hong Vu et al.).
- Papers incorporating deep learning-based NLP approaches, especially embedding techniques like Word2Vec, consistently demonstrate improved contextual understanding and higher predictive performance compared to classical feature extraction methods such as TF-IDF and BoW.
- The study by Dinh-Hong Vu et al. clearly shows that combining textual features with meta-information significantly enhances classification

performance, indicating that hybrid feature integration leads to better fraud detection robustness.

- Several papers (Aashir Amaar et al., Vidros et al., Shashikant Kumar et al.) emphasize that no single model is universally optimal; instead, performance heavily depends on dataset characteristics, class imbalance severity, feature engineering quality, and evaluation strategy.
- Class imbalance emerges as a recurring and critical challenge across the literature. Techniques such as ADASYN and SMOTE are used to artificially balance the dataset, demonstrating improved recall and F1-scores for fraudulent job detection, but also raising concerns about overfitting and synthetic bias.
- Behavioral and contextual textual features (job descriptions, company profiles, requirements) consistently contribute more to detection accuracy than simple metadata alone, proving that language patterns are central to recruitment fraud identification.
- Most studies rely heavily on internal evaluation metrics such as Accuracy, Precision, Recall, F1-score, ROC curve, and AUC. While these metrics allow algorithm comparison, they do not directly measure real-world deployment effectiveness or user-level impact.
- Several papers report very high accuracy scores (up to 99.9% in Aashir Amaar et al.), but such extremely high results raise questions about dataset generalization, oversampling bias, and robustness against unseen scam patterns.

- Scalability and real-time deployment remain underexplored areas. Although models are trained and evaluated experimentally, very few studies propose live, user-facing systems such as browser extensions or platform-level integration.
- Explainability and user trust mechanisms are largely absent in most studies, limiting transparency in decision-making when a job post is classified as fraudulent.
- Very few studies explore cross-platform or multi-source datasets beyond EMSCAD and Kaggle datasets, raising concerns about generalization across different countries, languages, and job market ecosystems.
- Most papers focus primarily on classification accuracy rather than integration into real-world systems such as job portals, Gmail scanning tools, LinkedIn integrations, or browser-based security tools.
- Overall, the literature demonstrates that NLP-based fake job detection is technically mature in terms of classification modeling, but still faces significant gaps in real-time deployment, scalability, cross-platform integration, generalization, and practical usability.
- These identified gaps clearly motivate the need for a real-time, browser-integrated NLP-based fake job detection system, such as the Chrome extension proposed in this project, which bridges the gap between high-accuracy research models and practical user-level fraud prevention.

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